



Newton-based matrix splitting method for generalized absolute value equation[☆]

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ABSTRACT

In this paper, based on the previous published work by Wang et al. (2019), by using the matrix splitting technique, Newton-based matrix splitting iterative method is established to solve the generalized absolute value equation. The proposed method not only covers the above modified Newton-type iterative method, but also generates some relaxation versions. Some convergence conditions of the proposed method with some special coefficient matrices are presented. The effectiveness and feasibility of the proposed method are confirmed by some numerical experiments.

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1. Introduction

As is known, for $A, B \in \mathbb{R}^{n \times n}$ and $b \in \mathbb{R}^n$, the generalized absolute value equation (GAVE) is to determine x such that

$$Ax - B|x| = b, \quad (1.1)$$

where $|\cdot|$ denotes the absolute value. Clearly, when $B = I$ in (1.1) with I denoting the identity matrix, the GAVE (1.1) becomes

$$Ax - |x| = b, \quad (1.2)$$

which is called the absolute value equation (AVE). The GAVE (1.1) implies that matrix B in (1.1) is nonzero, i.e., $B \neq 0$. Further, we know that the GAVE (1.1) is non-differentiable and nonlinear.

As a useful tool in the field of optimization, the GAVE (1.1) has received considerable attention because it arises in diverse fields, including linear programming, nonnegative constrained least squares problems, quadratic programming, complementarity problem, bimatrix games, and so on, see [1–6]. For instance, for obtaining the numerical solution of the well-known linear complementarity problem (LCP): determining z such that

$$Mz + q \geq 0, \quad z \geq 0 \text{ and } z^T(Mz + q) = 0, \text{ with } M \in \mathbb{R}^{n \times n}, q \in \mathbb{R}^n, \quad (1.3)$$

many efficient numerical methods can be designed by the form of the GAVE (1.1), such as the modulus-based matrix splitting (MMS) method [7] and its variants [8–14], and the generalized Newton method [15].

Investigating the structure of the GAVE (1.1), it is easy to see that the nonlinear and nondifferentiable term $B|x|$ often leads to the nondeterminacy of the solution of the GAVE (1.1). In this case, some theoretical results for the determinacy

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of the solution of the GAVE (1.1) have been obtained. For example, in [1], an alternative theorem for the GAVE (1.1) was given. From the perspective of the regularity of the interval matrix, the spectral radius and the singular value of the matrix, many sufficient conditions for the unique solvability of the GAVE (1.1) have been given in the literature [16–19]. Especially, in [20], Wu and Li presented some necessary and sufficient conditions for the unique solvability of the AVE (1.2), which have been extended to the GVAE (1.1) under certain conditions, see [21]. For other sufficient conditions for the unique solvability of the AVE (1.2), one can see [3,20].

In the last couple of years, under the existence and uniqueness of the solution of the GAVE (1.1) or the AVE (1.2) condition, to efficiently solve the GAVE (1.1) or the AVE (1.2), many efficient iterative methods have been designed, including the successive linearization method [2], the sign accord method [16], the search direction method [22], the Picard and Picard–HSS methods [17,23], and so on.

Recently, in [24], by expressing the AVE (1.2) as the following nonlinear equation

$$Ax - |x| - b = 0, \quad (1.4)$$

together with the generalized Jacobian $\partial|x|$ of $|x|$, substituting (1.4) into the well-known Newton iterative method

$$x^{k+1} = x^k - F'(x^k)^{-1}F(x^k) \quad (1.5)$$

leads to the generalized Newton (GN) method, where $F(x) = Ax - |x| - b$, which works as follows.

Method 1 (The GN Method). Assume that $x^0 \in \mathbb{R}^n$ is an arbitrary initial guess. For $k = 0, 1, 2, \dots$ until the iteration sequence $\{x^k\}_{k=0}^\infty \in \mathbb{R}^n$ is convergent, computing $x^{(k+1)}$ by

$$x^{k+1} = (A - D(x^k))^{-1}b, \quad k = 0, 1, \dots, \quad (1.6)$$

where $D(x^k) = \text{diag}(\text{sign}(x^k))$. Here, the vector $\text{sign}(x)$ consists of 1, 0, -1 dependent on whether the corresponding value of x is more than zero, equal to zero, less than zero.

Under appropriate hypotheses, Mangasarian [24] gave the convergence theory of the GN method and confirmed that the computational efficiency of the GN method overmatches the successive linearization method by some numerical experiments in [2]. Clearly, we use $BD(x^k)$ instead of $D(x^k)$ in (1.6), the corresponding GN method (1.6) can be used to solve the GAVE (1.1), see [25,26]. Whereas, it is noted that in the computation, the Jacobian matrix of the GN method is changed, the computations of the GN method can be much expensive. To avoid the changed Jacobian matrix, Wang et al. in [27] further exploited $F(x)$ and expressed $F(x)$ as

$$F(x) = S(x) + T(x),$$

where $S(x) = \Omega x + Ax$ and $T(x) = -\Omega x - B|x| - b$, together with the modified Newton-type iterative scheme in [28]

$$x^{k+1} = x^k - S'(x^k)^{-1}(S(x^k) + T(x^k)), \quad k = 0, 1, 2, \dots, \quad (1.7)$$

and designed the modified Newton (MN) method below.

Method 2 (The MN Method). Assume that $x^0 \in \mathbb{R}^n$ is an arbitrary initial guess. For $k = 0, 1, 2, \dots$ until the iteration sequence $\{x^k\}_{k=0}^\infty \in \mathbb{R}^n$ is convergent, computing $x^{(k+1)}$ by

$$x^{k+1} = (A + \Omega)^{-1}(\Omega x^k + B|x^k| + b), \quad k = 0, 1, 2, \dots, \quad (1.8)$$

where Ω is positive semi-definite.

Comparing the MN method with the GN method, a considerable advantage of the former consists in the fact that the changed coefficient matrix $A - D(x^k)$ is avoided. Whereas, a drawback of the MN method is that at very iteration it needs to tackle a linear system with the matrix $A + \Omega$. In particular, when the matrix $A + \Omega$ is ill-conditioned, the MN method may be more expensive in practical implementations. In addition, when the matrix A is indefinite, the MN method may be failure because the matrix $A + \Omega$ may be close to singular or be singular.

In this paper, assume that the GAVE (1.1) has a unique solution (see [21]), based on the separable property of the linear and the nonlinear terms, together with the Newton method and the matrix splitting technique, we establish a Newton-based matrix splitting iterative method for solving the GAVE (1.1). By using the matrix splitting suitably, the proposed method can include the Picard method [17] and the modified Newton method [27]. What is more, it also generates some relaxation versions. In particular, when matrix A in (1.1) is either an H_+ -matrix or a positive definite matrix, we give some convergence conditions of the proposed method. Further, we verify that the proposed method is feasible by some numerical examples.

The remainder of the paper is organized below. Some necessary terminology and useful lemmas are given in Section 2. In Section 3, Newton-based matrix splitting iterative method is established. In Section 4, the convergence properties of the proposed method are discussed in detail. In Section 5, the numerical results for the proposed method are reported. In Section 6, we draw some conclusions to finish this paper.

2. Preliminaries

This section consists of some necessary terminology and lemmas, which are suitable for the later discussion.

Let $A = (a_{ij}) \in \mathbb{R}^{n \times n}$. It is called as a Z -matrix if $a_{ij} \leq 0$ for $i \neq j$; a nonsingular M -matrix if $A^{-1} \geq 0$ and A is a Z -matrix; an H -matrix if its comparison matrix $\langle A \rangle = (\langle a \rangle_{ij}) \in \mathbb{R}^{n \times n}$ ($\langle a \rangle_{ii} = |a_{ii}|$ and $\langle a \rangle_{ij} = -|a_{ij}|$ for $i \neq j$) is a nonsingular M -matrix; an H_+ -matrix if A is an H -matrix with $\text{diag}(A) > 0$ in [29]; a positive definite matrix if $\frac{1}{2}(A^T + A)$ is positive definite; a strictly diagonally dominant (SDD) matrix if

$$|a_{ii}| > \sum_{j \neq i} |a_{ij}|, i = 1, 2, \dots, n,$$

see [30]. If $A \leq B$, with A and B being an M -matrix and a Z -matrix, respectively, then B is an M -matrix [30]. $\rho(A)$ and $\|A\|_2$, respectively, denote the spectral radius and the 2-norm of the matrix A .

Let $A = M - N$ with $\det(M) \neq 0$. Then it is named an H -compatible splitting if $\langle A \rangle = \langle M \rangle - |N|$ with $|N| = (|n_{ij}|)$; an H -splitting if $\langle M \rangle - |N|$ is a nonsingular M -matrix; an M -splitting if $N \geq 0$ and M is a nonsingular M -matrix. In addition, if $A = M - N$ is an M -splitting with A being a nonsingular M -matrix, then $\rho(M^{-1}N) < 1$.

Lemma 2.1 ([31]). If there exists $u \in \mathbb{R}^n$ with $u > 0$ such that $Au < u$ with $A \geq 0$, then $\rho(A) < 1$.

Lemma 2.2 ([32]). Let $A = D - B$, where $A \in \mathbb{R}^{n \times n}$ is an H -matrix and D is its diagonal part. Then matrices A and $|D|$ are nonsingular, $|A^{-1}| \leq \langle A \rangle^{-1}$ and $\rho(|D|^{-1}|B|) < 1$.

3. Newton-based matrix splitting method

In this section, Newton-based matrix splitting iteration method is established to solve the GAVE (1.1). To this end, we express the matrix A as $A = M - N$. Let

$$F(x) = \bar{S}(x) + \bar{T}(x), \quad (3.1)$$

where $\bar{S}(x) = \Omega x + Mx$ and $\bar{T}(x) = -\Omega x - Nx - B|x| - b$. Substituting (3.1) into (1.7) leads to a class of iteration methods, which is named as Newton-based matrix splitting iterative method and described below.

Method 3 (Newton-based Matrix Splitting Iterative Method). Assume that $x^0 \in \mathbb{R}^n$ is an arbitrary initial guess. For $k = 0, 1, 2, \dots$ until the iteration sequence $\{x^k\}_{k=0}^\infty \in \mathbb{R}^n$ is convergent, computing x^{k+1} by

$$\begin{aligned} x^{k+1} &= x^k - (M + \Omega)^{-1}(Ax^k - B|x^k| - b) \\ &= (M + \Omega)^{-1}((N + \Omega)x^k + B|x^k| + b), k = 0, 1, 2, \dots, \end{aligned} \quad (3.2)$$

where $\Omega + M$ is invertible and Ω is a given matrix.

Comparing Method 2 with Method 3, both need to tackle a linear system, but the coefficient matrix of the corresponding linear system is different. The former is matrix $A + \Omega$ and the latter is matrix $M + \Omega$. From the view of the computing efficiency, both may have been better than Method 1. It is because in actual computation, Method 1 needs to tackle the different linear system. Clearly, in Method 3, if we choose the suitable matrix Ω and matrix splitting of A , then it may be performed easily and may be fit for the indefinite matrix A as well. Furthermore, in actual implementations, we only consider to calculate the modulus of x^k rather than the projection.

It is noted that Method 3 for solving the GAVE (1.1) gives a general framework of Newton-based matrix splitting iterative method. Besides including the Picard method [17] and the MN method in [27] as two special cases, it can generate some Newton-based relaxation methods by selecting the matrix splitting suitably. Concretely, we split matrix A into three parts: $A = D - L - U$, where $D = \text{diag}(A)$, L and U , respectively, represent the strictly lower-triangular and upper-triangular part of A . Then

(a) for $M = A$ and $\Omega = N = 0$, Method 3 turns into the Picard method [17]

$$x^{k+1} = A^{-1}(B|x^k| + b);$$

(b) for $M = A$ and $N = 0$, Method 3 turns into the MN method [27]

$$x^{k+1} = (A + \Omega)^{-1}(\Omega x^k + B|x^k| + b);$$

(c) for $M = D$ and $N = L + U$, Method 3 becomes

$$x^{k+1} = (D + \Omega)^{-1}((\Omega + L + U)x^k + B|x^k| + b),$$

which can be called the Newton-based Jacobi (NJ) method;

(d) for $M = D - L$ and $N = U$, [Method 3](#) becomes

$$x^{k+1} = (D + \Omega - L)^{-1}((\Omega + U)x^k + B|x^k| + b),$$

which can be called the Newton-based Gauss-Seidel (NGS) method;

(e) for $M = \frac{1}{\alpha}D - L$ and $N = (\frac{1}{\alpha} - 1)D + U$, [Method 3](#) becomes

$$x^{k+1} = (D + \alpha\Omega - \alpha L)^{-1}((\alpha\Omega + (1 - \alpha)D + \alpha U)x^k + \alpha(B|x^k| + b)),$$

which can be called the Newton-based SOR (NSOR) method;

(f) for $M = \frac{1}{\alpha}(D - \beta L)$ and $N = \frac{1}{\alpha}((1 - \alpha)D + (\alpha - \beta)L + \alpha U)$, [Method 3](#) becomes

$$x^{k+1} = (D + \alpha\Omega - \beta L)^{-1}((\alpha\Omega + (1 - \alpha)D + (\alpha - \beta)L + \alpha U)x^k + \alpha(B|x^k| + b)),$$

which can be called the Newton-based AOR (NAOR) method;

(g) for $M = H$ and $N = -S$, where $H = \frac{1}{2}(A + A^T)$ and $S = \frac{1}{2}(A - A^T)$, [Method 3](#) becomes

$$x^{k+1} = (\Omega + H)^{-1}((\Omega - S)x^k + B|x^k| + b),$$

which can be called the Newton-based Hermitian and Skew-Hermitian (NHSS) method.

Comparing the Picard method and the MN method with the above-mentioned relaxation methods, the advantage of the latter is that at very iteration they are not involved with the computation of the inverse of the matrix A or $\Omega + A$. This implies that in the implementation, the efficiency of these Newton-based relaxation methods may be superior to the Picard method and the MN method.

4. Convergence property

In this section, we focus on discussing the convergence property of [Method 3](#). First, a general convergence property is given. Second, we give some sufficient conditions for [Method 3](#) when the matrix A in (1.1) is a positive definite matrix. Finally, we obtain some sufficient conditions for [Method 3](#) when the matrix A in (1.1) is an H_+ -matrix as well.

[Theorem 4.1](#) presents a general convergence condition of [Method 3](#) for solving the GAVE (1.1) when the related matrix is nonsingular.

Theorem 4.1. Let $A \in \mathbb{R}^{n \times n}$ and $A = M - N$ be its a splitting. Assume that the matrix $M + \Omega$ is invertible, where $\Omega \in \mathbb{R}^{n \times n}$ is given matrix. Define

$$f_\Omega = |(M + \Omega)^{-1}(N + \Omega)|, g_\Omega = |(M + \Omega)^{-1}B|.$$

If

$$\rho(f_\Omega + g_\Omega) < 1, \quad (4.1)$$

then [Method 3](#) is convergent.

Proof. Assume that a solution of the GVAE (1.1) is x^* . From (1.1), we have

$$Ax^* - B|x^*| = b,$$

which is equal to

$$(M + \Omega)x^* = (N + \Omega)x^* + B|x^*| + b, \quad (4.2)$$

Subtracting (4.2) from (3.2), we get

$$(M + \Omega)(x^{k+1} - x^*) = (N + \Omega)(x^k - x^*) + B(|x^k| - |x^*|).$$

Noticing that $M + \Omega$ is invertible, we get

$$x^{k+1} - x^* = (M + \Omega)^{-1}((N + \Omega)(x^k - x^*) + B(|x^k| - |x^*|)). \quad (4.3)$$

Using the absolute value for (4.3) leads to

$$\begin{aligned} |x^{k+1} - x^*| &= |(M + \Omega)^{-1}((N + \Omega)(x^k - x^*) + B(|x^k| - |x^*|))| \\ &= |(M + \Omega)^{-1}(N + \Omega)(x^k - x^*) + (M + \Omega)^{-1}B(|x^k| - |x^*|)| \\ &\leq |(M + \Omega)^{-1}(N + \Omega)(x^k - x^*)| + |(M + \Omega)^{-1}B(|x^k| - |x^*|)| \\ &\leq |(M + \Omega)^{-1}(N + \Omega)(x^k - x^*)| + |(M + \Omega)^{-1}B| \cdot \|x^k - x^*\| \end{aligned}$$

$$\begin{aligned} &\leq |(M + \Omega)^{-1}(N + \Omega)| \cdot |x^k - x^*| + |(M + \Omega)^{-1}B| \cdot |x^k - x^*| \\ &= (|(M + \Omega)^{-1}(N + \Omega)| + |(M + \Omega)^{-1}B|)|x^k - x^*|. \end{aligned}$$

Clearly, [Method 3](#) is convergent under the condition (4.1). $\square$

Since

$$|(M + \Omega)^{-1}(N + \Omega)| + |(M + \Omega)^{-1}B| \leq |(M + \Omega)^{-1}|(|N + \Omega| + |B|),$$

[Corollary 4.1](#) can be obtained.

Corollary 4.1. Let the conditions of [Theorem 4.1](#) be satisfied. If

$$\rho(|(M + \Omega)^{-1}|(|N + \Omega| + |B|)) < 1,$$

then [Method 3](#) is convergent.

Clearly, using $\|\cdot\|_2$ instead of $\rho(\cdot)$ in [Theorem 4.1](#) and [Corollary 4.1](#), the corresponding results are still valid. Combining the 2-norm version of the condition of [Corollary 4.1](#),

$$\|(M + \Omega)^{-1}\|_2(\|N + \Omega\|_2 + \|B\|_2) < 1,$$

with Banach perturbation Lemma in [33], [Corollary 4.2](#) can be given as well.

Corollary 4.2. Let A be positive definite and $A = M - N$ be its a splitting, where M is positive definite. Assume that the matrix $\Omega \in \mathbb{R}^{n \times n}$ is positive diagonal. If

$$\|M^{-1}\|_2 < \frac{1}{\|\Omega\|_2 + \|\Omega + N\|_2 + \|B\|_2}, \quad (4.4)$$

then [Method 3](#) is convergent.

Proof. According to the hypothesis, it is see to that matrix $M + \Omega$ is positive definite. Clearly, matrices $M + \Omega$ and M are invertible.

Based on the Banach perturbation lemma in [33], we have

$$\begin{aligned} \|(M + \Omega)^{-1}\|_2 &\leq \frac{\|M^{-1}\|_2}{1 - \|M^{-1}\|_2 \|\Omega\|_2} \\ &< \frac{1}{\frac{\|N + \Omega\|_2 + \|B\|_2 + \|\Omega\|_2}{\|M^{-1}\|_2}} \\ &= \frac{1}{\|N + \Omega\|_2 + \|B\|_2}. \end{aligned}$$

Therefore, [Method 3](#) is convergent under the condition (4.4). $\square$

When $A \in \mathbb{R}^{n \times n}$ is positive definite, for $\Omega = \omega I$ with $\omega > 0$, [Theorem 4.2](#) can be obtained.

Theorem 4.2. Let $A \in \mathbb{R}^{n \times n}$ be positive definite and $A = H + S$ be its a splitting with $H = \frac{1}{2}(A + A^T)$ and $S = \frac{1}{2}(A - A^T)$. Assume that $\|B\|_2 = \tau$ and $\Omega = \omega I$ with $\omega > 0$. If

$$\sqrt{\omega^2 + \mu_{\max}^2} < \omega + \lambda_{\min} - \tau, \quad (4.5)$$

where $\lambda_{\min}$ denotes the minimum eigenvalue of the matrix H and $\mu_{\max}$ denotes the maximum value of the absolute values of the eigenvalues of the matrix S , then [Method 3](#) is convergent.

Proof. Let $M = H$ and $N = -S$. Then

$$\begin{aligned} \|(M + \Omega)^{-1}(N + \Omega)\|_2 &= \|(\Omega + H)^{-1}(\Omega - S)\|_2 \\ &= \max_{\lambda_i \in \lambda(H), \mu_j \in \lambda(S)} \left| \frac{\sqrt{\omega^2 + \mu_j^2}}{\omega + \lambda_i} \right| \\ &= \frac{\sqrt{\omega^2 + \mu_{\max}^2}}{\omega + \lambda_{\min}} \end{aligned}$$

and

$$\|(M + \Omega)^{-1}B\|_2 = \|(H + \Omega)^{-1}B\|_2 \leq \|(H + \Omega)^{-1}\|_2 \|B\|_2 = \frac{\tau}{\omega + \lambda_{\min}}.$$

Hence,

$$\|(M + \Omega)^{-1}(N + \Omega)\|_2 + \|(M + \Omega)^{-1}B\|_2 \leq \frac{\sqrt{\omega^2 + \mu_{\max}^2}}{\omega + \lambda_{\min}} + \frac{\tau}{\omega + \lambda_{\min}}.$$

Using the 2-norm version of the condition of [Theorem 4.1](#) and straightforwardly simplifying the inequality

$$\frac{\sqrt{\omega^2 + \mu_{\max}^2}}{\omega + \lambda_{\min}} + \frac{\tau}{\omega + \lambda_{\min}} < 1,$$

we obtain that [Method 3](#) is convergent under the condition (4.5). $\square$

Remark 4.1. From the condition (4.5), it is easy to see that when $\lambda_{\min} > \tau + \mu_{\max}$, [Method 3](#) is convergent for $\omega > 0$.

When the matrix $A \in \mathbb{R}^{n \times n}$ is an H_+ -matrix, [Theorem 4.3](#) can be derived.

Theorem 4.3. Let $A = M - N$ be an H -splitting of the H_+ -matrix $A \in \mathbb{R}^{n \times n}$, and let Ω be a positive diagonal matrix. If

$$\rho((\Omega + \langle M \rangle)^{-1}(|N + \Omega| + |B|)) < 1, \quad (4.6)$$

then [Method 3](#) is convergent.

Proof. By assumptions and [Lemma 2.2](#), we know

$$|(\Omega + M)^{-1}| \leq (\langle M \rangle + \Omega)^{-1}.$$

Based on the proof of [Theorem 4.1](#), we have

$$\begin{aligned} |x^{k+1} - x^*| &\leq |(M + \Omega)^{-1}| \cdot (|N + \Omega| + |B|)|x^k - x^*| \\ &\leq (\langle M \rangle + \Omega)^{-1}(|N + \Omega| + |B|)|x^k - x^*|. \end{aligned}$$

Therefore, [Method 3](#) is convergent under the condition (4.6). $\square$

Further, if matrices A and Ω are the strictly diagonally dominant H_+ -matrices, then [Theorem 4.4](#) can be obtained.

Theorem 4.4. Let $\langle A \rangle = \langle M \rangle - |N|$, where $A \in \mathbb{R}^{n \times n}$ is an H_+ -matrix and M is strictly diagonally dominant. Assume that Ω is a strictly diagonally dominant H_+ -matrix. If

$$\rho((\langle M \rangle + \langle \Omega \rangle)^{-1}(|N + \Omega| + |B|)) < 1, \quad (4.7)$$

then [Method 3](#) is convergent.

Proof. Since $\langle A \rangle = \langle M \rangle - |N|$, clearly,

$$\langle A \rangle \leq \langle M \rangle \leq \text{diag}(M).$$

Naturally, matrix M is a strictly diagonally dominant H_+ -matrix. Further, we can obtain that $\langle M \rangle + \langle \Omega \rangle$ is an M -matrix because matrices M and Ω are the strictly diagonally dominant H_+ -matrices.

Let $M = (m_{ij}) \in \mathbb{R}^{n \times n}$ and $\Omega = (\omega_{ij}) \in \mathbb{R}^{n \times n}$. Then according to the following inequality

$$-|m_{ij} + \omega_{ij}| \geq -|m_{ij}| - |\omega_{ij}|,$$

we get

$$\langle M + \Omega \rangle \geq \langle M \rangle + \langle \Omega \rangle.$$

This shows that $\langle M + \Omega \rangle$ is an M -matrix and matrix $M + \Omega$ is an H_+ -matrix.

By proof of [Theorem 4.3](#) and [Lemma 2.2](#), we have

$$\begin{aligned} |x^{k+1} - x^*| &\leq \langle \Omega + M \rangle^{-1}(|N + \Omega| + |B|)|x^k - x^*| \\ &\leq (\langle \Omega \rangle + \langle M \rangle)^{-1}(|N + \Omega| + |B|)|x^k - x^*|. \end{aligned}$$

Therefore, [Method 3](#) is convergent under the condition (4.7). $\square$

Theorem 4.5. Let $A = D - \bar{B}$, where $A \in \mathbb{R}^{n \times n}$ is an H_+ -matrix and $D = \text{diag}(A)$. Let $\hat{B} = |\bar{B}| + |B|$ and $\rho := \rho(D^{-1}\hat{B})$. Assume that matrix Ω is positive diagonal. Then for any initial vector, the NAOR method converges for one of the following conditions:

(i) when $0 < \beta \leq \alpha$,

$$\rho < \frac{1 - |1 - \alpha|}{\alpha} \text{ and } 0 < \alpha < \frac{2}{1 + \rho};$$

(ii) when $\alpha < \beta$,

$$(2\beta - \alpha)\rho < \alpha < 2 - (2\beta - \alpha)\rho \text{ and } \rho < \frac{1 - |1 - \alpha|}{2\beta - \alpha}.$$

Proof. Let

$$M_\Omega = \Omega + \langle M \rangle, N_\Omega = |\Omega + N| + |B|, A_\Omega = M_\Omega - N_\Omega \text{ and } L_\Omega = M_\Omega^{-1}N_\Omega.$$

Then $\rho(L_\Omega) < 1$ with A_Ω being an M -matrix. Based on this, if $A = M - N$ is an M -matrix, where $M = \frac{1}{\alpha}(D - \beta L)$ is an M -matrix and

$$N = \frac{1}{\alpha}[(1 - \alpha)D + (\alpha - \beta)L + \alpha U] \geq 0,$$

then the NAOR method is convergent.

By the straightforward calculations, we have

$$M_\Omega = \Omega + \frac{1}{\alpha}(D - \beta|L|),$$

$$N_\Omega = \Omega + \frac{|1 - \alpha|}{\alpha}D + \frac{|\alpha - \beta|}{\alpha}|L| + |U| + |B|$$

and

$$A_\Omega = \frac{1 - |1 - \alpha|}{\alpha}D - \frac{|\alpha - \beta| + \beta}{\alpha}|L| - |U| - |B|. \quad (4.8)$$

Obviously, M_Ω is an M -matrix, $N_\Omega \geq 0$.

For $0 < \beta \leq \alpha$, from (4.8) we have

$$A_\Omega = \frac{1 - |1 - \alpha|}{\alpha}D - \hat{B}.$$

Clearly, if

$$1 - |1 - \alpha| > 0 \text{ and } \rho < \frac{1 - |1 - \alpha|}{\alpha}, \quad (4.9)$$

then A_Ω is an M -matrix. From (4.9), we have

$$\rho < \frac{1 - |1 - \alpha|}{\alpha}, 0 < \alpha < \frac{2}{1 + \rho}.$$

For $\beta > \alpha > 0$, from (4.8) we have

$$\begin{aligned} A_\Omega &= \frac{1 - |1 - \alpha|}{\alpha}D - \frac{2\beta - \alpha}{\alpha}|L| - |U| - |B| \\ &\geq \frac{1 - |1 - \alpha|}{\alpha}D - \frac{2\beta - \alpha}{\alpha}\hat{B} := K. \end{aligned}$$

Obviously, if

$$1 - |1 - \alpha| > 0 \text{ and } \rho < \frac{1 - |1 - \alpha|}{2\beta - \alpha}. \quad (4.10)$$

then K is an M -matrix. Further, we know that A_Ω is an M -matrix. From (4.10), we have

$$(2\beta - \alpha)\rho < \alpha < 2 - (2\beta - \alpha)\rho \text{ and } \rho < \frac{1 - |1 - \alpha|}{2\beta - \alpha}.$$

Thus, the results of Theorem 4.5 are valid. $\square$

Corollary 4.3. Under the conditions of Theorem 4.5, the following results are valid:

(1) the NSOR method is convergent for

$$0 < \alpha < \frac{2}{1 + \rho} \text{ and } \rho < \frac{1 - |1 - \alpha|}{\alpha};$$

(2) the NGS method is convergent for $\rho < 1$;

(3) the NJ method is convergent for $\rho < 1$.

Similarly, let

$$M_\Omega = \langle \Omega \rangle + \langle M \rangle, N_\Omega = |\Omega| + |N| + |B|,$$

where

$$M = \frac{1}{\alpha}(D - \beta L) \text{ and } N = \frac{1}{\alpha}[(1 - \alpha)D + (\alpha - \beta)L + \alpha U].$$

Then Corollary 4.4 can be obtained. The proof of Corollary 4.4 is omitted.

Corollary 4.4. Let $A = D - \bar{B}$, where $A \in \mathbb{R}^{n \times n}$ is an H_+ -matrix and $D = \text{diag}(A)$. Assume that matrix Ω is a strictly diagonally dominant H_+ -matrix. Let $\hat{B} = |\bar{B}| + |B| + |\Omega| - \langle \Omega \rangle$ and $\rho := \rho(D^{-1}\hat{B})$. Then for any initial vector, the NAOR method converges for one of the following conditions:

(i) when $0 < \beta \leq \alpha$,

$$\rho < \frac{1 - |1 - \alpha|}{\alpha} \text{ and } 0 < \alpha < \frac{2}{1 + \rho};$$

(ii) when $\alpha < \beta$,

$$(2\beta - \alpha)\rho < \alpha < 2 - (2\beta - \alpha)\rho \text{ and } \rho < \frac{1 - |1 - \alpha|}{2\beta - \alpha}.$$

Theorem 4.6. Let Ω be a positive diagonal matrix. Let H_+ -matrix $A \in \mathbb{R}^{n \times n}$ be with $A = D - L - U = D + \bar{B}$ such that $\langle A \rangle = D - |L| - |U|$. Let $\hat{B} = |\bar{B}| + |B|$ and $\rho := \rho(D^{-1}\hat{B}) < 1$. If

$$\max\{\alpha, 2\beta - \alpha\}\rho < \min\{\alpha, 2 - \alpha\}. \quad (4.11)$$

then the NAOR method is convergent.

Proof. Let

$$W = (\langle M \rangle + \Omega)^{-1}(|N + \Omega| + |B|),$$

where

$$M = \frac{1}{\alpha}(D - \beta L) \text{ and } N = \frac{1}{\alpha}((1 - \alpha)D + (\alpha - \beta)L + \alpha U).$$

Then we only prove that $\rho(W) < 1$. Since $1 - |1 - \alpha| = \min\{\alpha, 2 - \alpha\}$ and

$$\begin{aligned} |\alpha\bar{B} - \beta L| + \beta|L| + \alpha|B| &\leq \max\{\alpha, 2\beta - \alpha\}(|\bar{B}| + |B|) \\ &= \max\{\alpha, 2\beta - \alpha\}\hat{B}, \end{aligned}$$

then

$$\begin{aligned} W &= (\langle M \rangle + \Omega)^{-1}(|N + \Omega| + |B|) \\ &= \left(\frac{1}{\alpha}\langle D - \beta L \rangle + \Omega\right)^{-1}\left(\frac{1}{\alpha}((1 - \alpha)D + (\alpha - \beta)L + \alpha U) + \Omega + |B|\right) \\ &= (D + \alpha\Omega - \beta|L|)^{-1}((1 - \alpha)D + (\alpha - \beta)L + \alpha U + \alpha\Omega + \alpha|B|) \\ &= I - (D + \alpha\Omega - \beta|L|)^{-1}((1 - |1 - \alpha|)D - |\alpha\bar{B} - \beta L| - \beta|L| - \alpha|B|) \\ &\leq \tilde{U}, \end{aligned}$$

where $\tilde{U} = I - (\alpha\Omega + D - \beta|L|)^{-1}D(\min\{2 - \alpha, \alpha\}I - \max\{\alpha, 2\beta - \alpha\}D^{-1}\hat{B})$.

Noting that $\rho(D^{-1}\hat{B}) < 1$, there is an arbitrary small $\varepsilon > 0$ such that

$$\rho_\varepsilon := \rho(J_\varepsilon) < 1,$$

where $J_\varepsilon := D^{-1}\hat{B} + \varepsilon e e^T \geq 0$ and $e := (1, \dots, 1)^T \in \mathbb{R}^n$. Using Perron–Frobenius theorem in [30], there is a positive vector $v_\varepsilon \in \mathbb{R}^n$ such that

$$J_\varepsilon v_\varepsilon = \rho_\varepsilon v_\varepsilon.$$

Further,

$$\begin{aligned} \tilde{U}v_\varepsilon &\leq v_\varepsilon - (\alpha\Omega + D - \beta|L|)^{-1}D(\min\{2 - \alpha, \alpha\}v_\varepsilon - \max\{\alpha, 2\beta - \alpha\}J_\varepsilon v_\varepsilon) \\ &= v_\varepsilon - (\min\{2 - \alpha, \alpha\} - \max\{\alpha, 2\beta - \alpha\}\rho_\varepsilon)(\alpha\Omega + D - \beta|L|)^{-1}Dv_\varepsilon \\ &\leq v_\varepsilon - (\min\{2 - \alpha, \alpha\} - \max\{\alpha, 2\beta - \alpha\}\rho_\varepsilon)(\alpha\Omega + D)^{-1}Dv_\varepsilon \\ &< v_\varepsilon. \end{aligned}$$

From Lemma 2.1, we know that $\rho(W) \leq \rho(\tilde{U}) < 1$. This implies that the NAOR method converges. $\square$

Remark 4.2. The convergence conditions of [Theorem 4.6](#) for the NAOR method are without one parameter being greater than the other. That is to say, the relaxation convergence area is provided by [Theorem 4.6](#), which chooses two parameters α and β freely in a way.

Corollary 4.5. Under the conditions of [Theorem 4.6](#), the following results are valid:

(1) the NSOR method is convergent for

$$\rho < \min\left\{1, \frac{2-\alpha}{\alpha}\right\};$$

(2) the NGS method is convergent for $\rho := \rho(D^{-1}\hat{B}) < 1$;

(3) the NJ method is convergent for $\rho := \rho(D^{-1}\hat{B}) < 1$.

Corollary 4.6. Let Ω be a strictly diagonally dominant H_+ -matrix. Let $A \in \mathbb{R}^{n \times n}$ be a strictly diagonally dominant H_+ -matrix such that $\langle A \rangle = D - |L| - |U|$. Let $\hat{B} = |\hat{B}| + |B| + |\Omega| - \langle \Omega \rangle$ and $\rho := \rho(D^{-1}\hat{B}) < 1$ with $\hat{B} = D - A$ and $D = \text{diag}(A)$. If

$$\max\{\alpha, 2\beta - \alpha\}\rho < \min\{\alpha, 2 - \alpha\},$$

then the NAOR method is convergent.

Remark 4.3. Clearly, the above convergence conditions can be extended to the AVE [\(1.2\)](#).

5. Numerical experiments

This section involves four examples to show the performance of the proposed method for solving the GAVE [\(1.1\)](#) from three aspects: the iteration count (indicated as 'IT'), the elapsed CPU time (indicated as 'CPU') and the relative residual error (indicated as 'RES'). Here, 'RES' is set to be

$$\text{RES} = \|Ax^k - B|x^k| - b\|_2 / \|b\|_2.$$

In our computations, the choice of the initial vector of the tested methods is

$$x^0 = (1, 0, 1, 0, \dots, 1, 0, \dots)^T \in \mathbb{R}^T.$$

and all iterations are stopped once $\text{RES} < 10^{-6}$ or the prescribed iteration count $k_{\max} = 500$ is surpassed. In the table, '—' denotes the iteration counts larger than 500 steps or the CPU times larger than 500 s. All the tests are operated on MATLAB 7.0.

As is known, using $w = |x| + x$ and $z = |x| - x$ for the LCP [\(1.3\)](#) leads to the following GAVE

$$Ax - B|x| = q, \tag{5.1}$$

where $A = M + I$, $B = M - I$ and $x = \frac{1}{2}((M - I)z + q)$. Based on this, we consider the following examples.

Example 1 ([\[27\]](#)). Consider the LCP(q, M), where $M = \hat{M} + \mu I$ with

$$\hat{M} = \text{tridiag}(-I, S, -I) \in \mathbb{R}^{n \times n} \text{ and } S = \text{tridiag}(-1, 4, -1) \in \mathbb{R}^{m \times m},$$

and $q = -Mz^*$ with $z^* = (1.2, 1.2, \dots, 1.2, \dots)^T \in \mathbb{R}^n$ being the unique solvability of the LCP(q, M). In this case, the unique solvability of the corresponding GAVE [\(5.1\)](#) is $x^* = (-0.6, -0.6, \dots, -0.6, \dots)^T \in \mathbb{R}^n$.

For [Example 1](#), based on the different problem sizes of n , we test the Picard method, the MN method, the NJ method, the NGS method and the NSOR method. Here, for these five tested methods, the choice of Ω is $\Omega = \hat{M}$, $1.5\hat{M}$. Using these five methods, we need to deal with the linear systems with the matrices A , $A + \Omega$, $D + \Omega$, $D + \Omega - L$ and $D + \alpha\Omega - \alpha L$, respectively. In the implementations, for solving these linear systems, the sparse LU factorization or the sparse Cholesky factorization can be employed. To improve the rate of convergence of the NSOR method, the choice of the parameter α is the experimentally optimal parameter $\alpha_{\exp}$ such that the NSOR method has the smallest iteration step.

For [Example 1](#), to show the advantages of these relaxation versions, we take two values for the parameter μ : $\mu = 4$ and $\mu = -1$. Under the circumstances, the matrix A is symmetric positive definite.

[Tables 1](#) and [2](#) list the numerical results of these five tested methods for $\mu = 4$. The numerical results in [Tables 1](#) and [2](#) show that these five tested methods can fast converge to the solution x^* for the different problem sizes of n . Not only that, all of these five tested methods are relative stable. This implies that their iteration counts are hardly sensitive to change in the mesh sizes.

Further, the numerical results in [Tables 1](#) and [2](#) show that the NSOR method requires the least iteration counts and the Picard method requires the maximum iteration counts. In terms of the elapsed CPU times, the NSOR method requires the least computing times. That is to say, when the experimentally optimal parameter $\alpha_{\exp}$ is used, the NSOR method is the most effective, compared with the other four methods.

Table 1Numerical comparison of [Example 1](#) for $\mu = 4$ with $\Omega = \hat{M}$.

	n	3600	4900	6400	8100	10 000
Picard	IT	64	63	63	63	63
	CPU	0.266	0.391	0.594	0.828	1.094
	RES	8.9163e-7	9.9983e-7	9.4233e-7	8.9429e-7	8.5340e-7
MN	IT	24	24	24	24	24
	CPU	0.109	0.11	0.172	0.234	0.297
	RES	6.4043e-7	6.6143e-7	6.7725e-7	6.8960e-7	6.9950e-7
NJ	IT	12	12	12	12	12
	CPU	0.062	0.094	0.141	0.171	0.266
	RES	8.5417e-7	7.9553e-7	7.4759e-7	7.0745e-7	6.7322e-7
NGS	IT	11	11	11	11	11
	CPU	0.047	0.079	0.109	0.14	0.156
	RES	3.5695e-7	3.4869e-7	3.4224e-7	3.3705e-7	3.3279e-7
NSOR	α_{exp}	0.9	0.9	0.9	0.9	0.9
	IT	9	9	9	9	9
	CPU	0.047	0.063	0.078	0.125	0.14
	RES	1.1928e-7	1.8941e-7	1.8663e-7	1.8440e-7	1.8257e-7

Table 2Numerical comparison of [Example 1](#) for $\mu = 4$ with $\Omega = 1.5\hat{M}$.

	n	3600	4900	6400	8100	10 000
Picard	IT	64	63	63	63	63
	CPU	0.266	0.391	0.594	0.828	1.094
	RES	8.9163e-7	9.9983e-7	9.4233e-7	8.9429e-7	8.5340e-7
MN	IT	24	24	24	24	24
	CPU	0.11	0.114	0.175	0.249	0.322
	RES	5.9923e-7	6.2604e-7	6.4623e-7	6.6199e-7	6.7462e-7
NJ	IT	8	8	8	8	8
	CPU	0.062	0.086	0.137	0.17	0.21
	RES	8.5417e-7	7.9553e-7	7.4759e-7	7.0745e-7	6.7322e-7
NGS	IT	8	8	8	8	8
	CPU	0.037	0.05	0.075	0.093	0.124
	RES	1.7936e-7	1.6959e-7	1.6180e-7	1.5543e-7	1.5011e-7
NSOR	α_{exp}	0.9	0.9	0.9	0.9	0.9
	IT	6	6	6	6	6
	CPU	0.032	0.048	0.058	0.084	0.109
	RES	6.5242e-7	5.9361e-7	5.4778e-7	5.1085e-7	4.8032e-7

Table 3Numerical comparison of [Example 1](#) for $\mu = -1$ with $\Omega = \hat{M}$.

	n	3600	4900	6400	8100	10 000
Picard	IT	–	–	–	–	–
	CPU	–	–	–	–	–
	RES	–	–	–	–	–
MN	IT	–	–	–	–	–
	CPU	–	–	–	–	–
	RES	–	–	–	–	–
NJ	IT	51	51	50	50	50
	CPU	0.204	0.296	0.407	0.563	0.688
	RES	8.7996e-7	8.2701e-7	9.7990e-7	9.3796e-7	9.0284e-7
NGS	IT	59	58	58	57	57
	CPU	0.359	0.531	0.734	1.031	1.36
	RES	8.5017e-7	9.3343e-7	8.7524e-7	9.7771e-7	9.2895e-7
NSOR	α_{exp}	1.33	1.33	1.32	1.31	1.3
	IT	54	53	53	53	53
	CPU	0.218	0.328	0.437	0.578	0.766
	RES	8.6010e-7	9.6545e-7	9.2671e-7	8.9233e-7	8.5697e-7

Tables 3 and 4 present the numerical results of these five tested methods for $\mu = -1$. The numerical results of Tables 3 and 4 show that the Picard method and the MN method are not convergent when $\mu = -1$. Whereas, the NJ method, the

Table 4
Numerical comparison of [Example 1](#) for $\mu = -1$ with $\Omega = 1.5\hat{M}$.

	n	3600	4900	6400	8100	10 000
Picard	IT	–	–	–	–	–
	CPU	–	–	–	–	–
	RES	–	–	–	–	–
MN	IT	–	–	–	–	–
	CPU	–	–	–	–	–
	RES	–	–	–	–	–
NJ	IT	68	68	67	67	67
	CPU	0.292	0.362	0.555	0.728	0.999
	RES	9.2712	8.6413	9.4917	9.0041	8.5918
NGS	IT	76	75	75	74	74
	CPU	0.676	0.967	1.472	1.924	2.139
	RES	9.1417	9.6609	9.0598	9.7385	9.2533
NSOR	α_{exp}	1.3	1.3	1.3	1.3	1.3
	IT	71	71	70	70	69
	CPU	0.31	0.37	0.562	0.771	1.022
	RES	9.6842	9.001	9.6678	9.1303	9.9378

Table 5
Numerical comparison of [Example 2](#) for $\mu = 4$ with $\Omega = \hat{M}$.

	n	3600	4900	6400	8100	10 000
Picard	IT	65	65	65	64	64
	CPU	0.266	0.391	0.593	0.812	1.093
	RES	9.8655e–7	9.2007e–7	8.6567e–7	9.7913e–7	9.3294e–7
MN	IT	23	23	24	24	24
	CPU	0.094	0.11	0.172	0.234	0.297
	RES	9.4471e–7	9.9690e–7	6.1709e–7	6.3602e–7	6.5121e–7
NJ	IT	13	13	13	13	13
	CPU	0.047	0.078	0.109	0.156	0.188
	RES	5.1537e–7	4.7971e–7	4.5055e–7	4.2614e–7	4.0530e–7
NGS	IT	13	13	13	13	13
	CPU	0.047	0.078	0.109	0.141	0.188
	RES	3.8144e–7	3.8683e–7	3.9089e–7	3.9404e–7	3.9657e–7
NSOR	α_{exp}	0.8	0.8	0.8	0.8	0.8
	IT	7	7	7	7	7
	CPU	0.047	0.063	0.094	0.125	0.172
	RES	1.0295e–7	1.0036e–7	9.8329e–8	9.6695e–8	9.5351e–8

NGS method and the NSOR method are convergent. This implies that these relaxation versions compared to the Picard method and the MN method may be competitive. That is to say, the Newton-based relaxation methods are superior to the Picard method and the MN method under certain conditions.

Example 2 ([27]). Consider the $LCP(q, M)$, where $M = \hat{M} + \mu I$ with

$$\hat{M} = \text{tridiag}(-1.5I, S, -0.5I) \in \mathbb{R}^{n \times n} \text{ and } S = \text{tridiag}(-1.5, 4, -0.5) \in \mathbb{R}^{m \times m},$$

and $q = -Mz^*$ with $z^* = (1.2, 1.2, \dots, 1.2, \dots)^T \in \mathbb{R}^n$ being the unique solvability of the $LCP(q, M)$. In this case, the unique solvability of the corresponding GAVE (5.1) is $x^* = (-0.6, -0.6, \dots, -0.6, \dots)^T \in \mathbb{R}^n$.

For [Example 2](#), to improve the convergence speed of all the tested methods, the choice of Ω is $\Omega = \hat{M}, 1.2\hat{M}$. For the different problem sizes of n , the iteration counts, the CPU times, and the residual errors for [Example 2](#) with $\mu = 4$ are listed in [Tables 5](#) and [6](#). In this case, the matrix A is a strictly diagonally dominant H_+ -matrix. The numerical results in [Tables 5](#) and [6](#) further confirm the observations obtained from [Tables 1](#) and [2](#), i.e., the NSOR method is the best one and its computing efficiency is superior to the other four tested methods.

Similarly, in [Tables 7](#) and [8](#), we also list the iteration counts, the CPU times, and the residual errors for [Example 2](#) with $\mu = -1$ for the different problem sizes of n . It is noted that the matrix A is an irreducible and weakly diagonally dominant H_+ -matrix when $\mu = -1$. The numerical results in [Tables 7](#) and [8](#) show that the Picard method and the MN method do not converge indeed for $\mu = -1$. Whereas, the NJ method, the NGS method and the NSOR method are convergent. This conclusion is in line with the observations obtained from [Tables 3](#) and [4](#). The numerical results in [Tables 7](#) and [8](#) still confirm that the Newton-based relaxation methods are superior to the Picard method and the MN method under certain conditions.

Table 6
Numerical comparison of Example 2 for $\mu = 4$ with $\Omega = 1.2\hat{M}$.

	n	3600	4900	6400	8100	10 000
Picard	IT	65	65	65	64	64
	CPU	0.322	0.394	0.549	0.754	1.027
	RES	9.8655	9.2007	8.6567	9.7913	9.3294
MN	IT	23	23	23	24	24
	CPU	0.084	0.106	0.15	0.213	0.287
	RES	8.7540	9.3733	9.8400	6.0710	6.2515
NJ	IT	9	9	9	9	9
	CPU	0.052	0.066	0.093	0.126	0.172
	RES	8.3621	8.1496	7.9831	7.8490	7.7388
NGS	IT	13	13	13	13	13
	CPU	0.058	0.079	0.112	0.156	0.212
	RES	3.5226	3.6155	3.6855	3.7400	3.7838
NSOR	α_{exp}	0.82	0.8	0.8	0.79	0.82
	IT	6	6	6	6	6
	CPU	0.042	0.049	0.07	0.097	0.131
	RES	1.7938	2.5895	2.4580	8.7658	8.7424

Table 7
Numerical comparison of Example 2 for $\mu = -1$ with $\Omega = \hat{M}$.

	n	3600	4900	6400	8100	10 000
Picard	IT	–	–	–	–	–
	CPU	–	–	–	–	–
	RES	–	–	–	–	–
MN	IT	–	–	–	–	–
	CPU	–	–	–	–	–
	RES	–	–	–	–	–
NJ	IT	144	142	140	138	136
	CPU	0.672	0.953	1.36	1.828	2.453
	RES	9.7797e–7	9.5447e–7	9.5175e–7	9.6464e–7	9.9032e–7
NGS	IT	108	106	105	103	102
	CPU	0.578	0.813	1.125	1.484	1.891
	RES	9.3164e–7	9.6558e–7	9.2823e–7	9.9901e–7	9.8872e–7
NSOR	α_{exp}	1.1	1.08	1.08	1.08	1.08
	IT	97	98	96	95	95
	CPU	0.406	0.578	0.796	1.047	1.344
	RES	9.6696e–7	9.1422e–7	9.8778e–8	9.8099e–8	9.0054e–8

Table 8
Numerical comparison of Example 2 for $\mu = -1$ with $\Omega = 1.2\hat{M}$.

	n	3600	4900	6400	8100	10 000
Picard	IT	–	–	–	–	–
	CPU	–	–	–	–	–
	RES	–	–	–	–	–
MN	IT	–	–	–	–	–
	CPU	–	–	–	–	–
	RES	–	–	–	–	–
NJ	IT	151	149	147	145	143
	CPU	0.916	1.331	1.949	2.658	3.722
	RES	9.7548	9.4768	9.4064	9.4902	9.6982
NGS	IT	115	113	112	111	109
	CPU	0.654	0.798	1.132	1.593	2.049
	RES	9.5611	9.8163	9.3920	9.1337	9.8627
NSOR	α_{exp}	1.134	1.09	1.081	1.08	1.051
	IT	101	104	103	102	104
	CPU	0.46	0.554	0.894	1.11	1.528
	RES	9.9591	9.4805	9.9844	9.8923	9.5983

Numerical results of Examples 1 and 2 show that the Newton-based matrix splitting iterative method is superior to the Picard method and the MN method under certain conditions. Particularly, the NSOR method performs the best in terms of the computational efficiency, compared with the other four tested methods. Next, we use the Newton-based matrix

Table 9
Numerical comparison of Example 3 with $\Omega = 0.1\text{diag}(A)$.

	n	8000	9000	10000	11000
NJ	IT	22	22	22	22
	CPU	0.044	0.047	0.053	0.063
	RES	7.6426e-7	7.6426e-7	7.6426e-7	7.6426e-7
MGS	IT	16	16	16	16
	CPU	0.039	0.044	0.049	0.053
	RES	5.6590e-7	5.6530e-7	5.6483e-7	5.6444e-7
NSOR	α_{exp}	1.2	1.2	1.2	1.2
	IT	11	11	11	11
	CPU	0.031	0.032	0.042	0.046
	RES	6.3948e-7	6.2588e-7	6.1478e-7	6.0555e-7

Table 10
Numerical comparison of Example 3 with $\Omega = 0.5\text{diag}(A)$.

	n	8000	9000	10000	11000
NJ	IT	33	33	33	33
	CPU	0.064	0.079	0.135	0.14
	RES	7.2850e-7	7.2849e-7	7.2848e-7	7.2848e-7
MGS	IT	27	27	27	27
	CPU	0.061	0.07	0.078	0.083
	RES	6.8958e-7	6.8936e-7	6.8919e-7	6.8905e-7
NSOR	α_{exp}	2.2	2.2	2.2	2.2
	IT	11	11	11	11
	CPU	0.031	0.039	0.043	0.046
	RES	5.5272e-7	5.2508e-7	5.0187e-7	4.8205e-7

splitting iterative method for solving the GAVE (1.1), which is from the Black–Scholes American option pricing and least squares problem.

Example 3 (Black–Scholes American Option Pricing). We consider the Black–Scholes American option pricing in [34]. Finding the price $u(x, t)$ of American put options such that

$$\frac{\partial u}{\partial t} - \frac{\partial^2 u}{\partial x^2} \geq 0, u(x, t) - g(x, t) \geq 0, \left(\frac{\partial u}{\partial t} - \frac{\partial^2 u}{\partial x^2}\right)(u(x, t) - g(x, t)) = 0, \quad (5.2)$$

with $u(x, 0) = g(x, 0)$, $\lim_{x \rightarrow \pm\infty} u(x, t) = \lim_{x \rightarrow \pm\infty} g(x, t)$, $(x, t) \in (-\infty, +\infty) \times [0, T]$ and $x \in [a, b]$, where the values of a and b are chosen in terms of the approach in [34]. To discretize (5.2), we adopt the forward difference scheme for time t and implicit difference scheme for the price x , and obtain the linear complementarity problem

$$w := Mz - q \geq 0, z - g \geq 0 \text{ and } w^T(z - g) = 0, \quad (5.3)$$

with $M = \text{tridiag}(-\tau, 1 + 2\tau, -\tau)$ and $\tau = \frac{\Delta t}{(\Delta x)^2}$, where Δt , Δx denote the time step and the price step, respectively. In our numerical computations, we take $g = 0.5z^*$, and $z^* = (1, 0, 1, 0, \dots, 1, 0, \dots)^T$. The vector q is to be adjusted such that $q = Mz^* - w^*$, where $w^* = (0, 1, 0, 1, \dots, 0, 1, \dots)^T$.

Based on the approach of Examples 1 and 2, for Example 3, we first translate the LCP (5.3) into the GAVE (5.1), and then use the NJ method, the NGS method and the NSOR method for solving the corresponding generalized absolute value equations.

In our computations, for the NJ method, the NGS method and the NSOR method, we consider two cases of Ω : $\Omega = 0.1\text{diag}(A)$, $\Omega = 0.5\text{diag}(A)$ with $A = M + I$. In this case, Tables 9 and 10 list the numerical results (including IT, CPU and RES) for these three tested methods for Example 3 with the different problem sizes of n and $\tau = 1$. These numerical results in Tables 9 and 10 draw the same conclusion as before that the NSOR method performs the best, compared with the NJ method and the NGS method under certain conditions.

Example 4 (Least Squares Problem). We use the matrices ash608, ash958, illc1033, well1033 from the UF Sparse Matrix Collection,¹ which is from least squares problem. Generic properties of the test matrix are given in Table 11.

For Example 4, we first equivalently translate least squares problem into the form of the LCP (1.3), see [[35], p. 262], then translate the corresponding LCP into the GAVE (5.1), and finally use the NJ method, the NGS method and the NSOR method for solving the corresponding generalized absolute value equations.

¹ <https://www.cise.ufl.edu/research/sparse/matrices>.

Table 11
Matrix properties for [Example 4](#).

Matrix	n	m	$\text{nnz}(A)$
ash608	608	188	1216
ash958	958	292	1916
illc1033	1033	320	4719
well1033	1033	320	4732

Table 12
Numerical comparison of [Example 4](#) with $\Omega = I$.

	Matrix	ash608	ash958	illc1033	well1033
NJ	IT	83	79	13	13
	CPU	0.031	0.032	0.016	0.016
	RES	8.7563e-7	9.0360e-7	3.5413e-7	7.8939e-7
MGS	IT	95	90	18	16
	CPU	0.047	0.043	0.016	0.016
	RES	8.8301e-7	9.0661e-7	8.9581e-7	6.8380e-7
NSOR	α_{exp}	5.2	5.6	2	1.6
	IT	34	33	5	11
	CPU	0.015	0.015	0.015	0.015
	RES	9.2992e-7	8.4469e-7	1.1771e-12	2.7696e-7

Table 13
Numerical comparison of [Example 4](#) with $\Omega = 2I$.

	Matrix	ash608	ash958	illc1033	well1033
NJ	IT	89	85	20	21
	CPU	0.031	0.032	0.016	0.016
	RES	8.9810e-7	9.2824e-7	7.6057e-7	6.2986e-7
MGS	IT	101	97	27	24
	CPU	0.047	0.043	0.031	0.034
	RES	9.7341e-7	8.8687e-7	7.4388e-7	6.8706e-7
NSOR	α_{exp}	16	21	5.5	4.3
	IT	33	32	13	12
	CPU	0.015	0.015	0.015	0.015
	RES	8.4272e-7	8.5385e-7	2.3677e-7	4.5864e-7

In our computations, for the NJ method, the NGS method and the NSOR method, we consider two cases of Ω : $\Omega = I$, $\Omega = 2I$. In this case, [Tables 12](#) and [13](#) list some numerical results (including IT, CPU and RES) for these three tested methods for [Example 4](#). These numerical results in [Tables 12](#) and [13](#) still draw the same conclusion as before that the NSOR method performs the best, compared with the NJ method and the NGS method under certain conditions.

6. Conclusions

In this paper, by using the separable property of the linear and the nonlinear terms of the GAVE (1.1), together with the Newton method and the matrix splitting technique, Newton-based matrix splitting iterative method has been designed to solve the GAVE (1.1). When the system matrix A of the GAVE (1.1) is a positive definite matrix or an H_+ -matrix, we obtain some convergence conditions of the proposed method. The proposed method may be more efficient when the suitable parameter matrix is chosen. Numerical experiments verify that some relaxation versions of the proposed method are superior to the Picard method and the MN method under certain conditions.

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